Lab 9: Animation and Collision

Computer Graphics Lab, Wolfram Mathematica

In[1]:=
```
SetOptions[EvaluationNotebook[], ShowCellLabel → True];
```

In[2]:=
```
SetOptions[{Plot, Plot3D, ContourPlot, ContourPlot3D, DensityPlot,
    ParametricPlot, ParametricPlot3D, ListPlot, ListLinePlot, ListPointPlot3D,
    Graphics, Graphics3D, PolarPlot}, ImageSize → Small, AxesLabel → {"x", "y", "z"}];
```

In[3]:=
```
(* Problem 1 *)
```

In[4]:=
```
SP2[t_] := 0.15 {(3 / 4) * (t) * (t^2 - 11),
    t^4 - 12 * t^2, (1 / 30) * (t) * (t^2 - 1) * (t^2 - 3) * (t^2 - 9) * (t^2 - 12)}
```

In[5]:=
```
SetDirectory[NotebookDirectory[]];
```

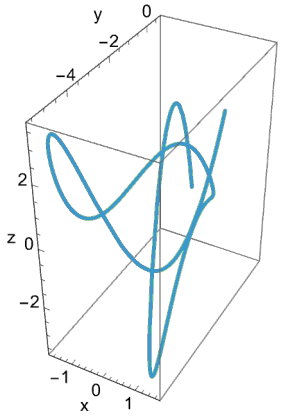
In[6]:=
```
D[SP2[t], t]
```

Out[6]=
$$\{0.225\,t^2 + 0.1125\,\left(-11 + t^2\right),\ 0.15\,\left(-24\,t + 4\,t^3\right),\ 0.01\,t^2\,\left(-12 + t^2\right)\,\left(-9 + t^2\right)\,\left(-3 + t^2\right) + 0.01\,t^2\,\left(-12 + t^2\right)\,\left(-9 + t^2\right)\,\left(-1 + t^2\right) + 0.01\,t^2\,\left(-12 + t^2\right)\,\left(-3 + t^2\right)\,\left(-1 + t^2\right) + 0.01\,t^2\,\left(-9 + t^2\right)\,\left(-3 + t^2\right)\,\left(-1 + t^2\right) + 0.005\,\left(-12 + t^2\right)\,\left(-9 + t^2\right)\,\left(-3 + t^2\right)\,\left(-1 + t^2\right)\}$$

In[7]:=
```
dSP2[t_] := Evaluate[D[SP2[t], t]]
```

In[8]:=
```
Traj2 = ParametricPlot3D[SP2[t], {t, -3.5, 3.5}]
```

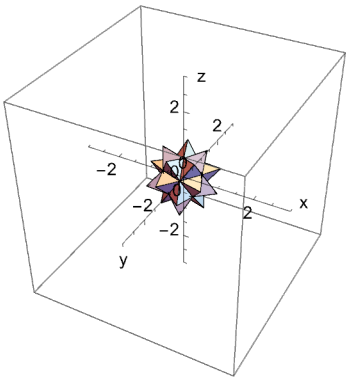
Out[8]=



In[9]:=
```
SpikeyT2 = Import["spikeymodel.stl", "PolygonObjects"];
```

In[10]:=
```
GSpiketT2 = Graphics3D[SpikeyT2, Axes → True, PlotRange → 3, AxesOrigin → {0, 0, 0}]
```

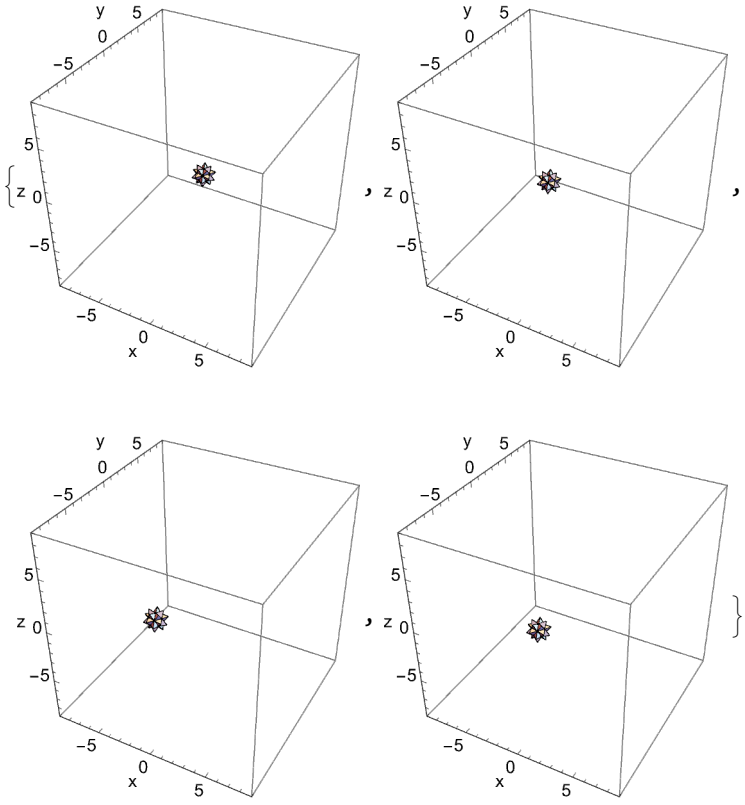
Out[10]=



In[11]:=
```
PositionT2[P_] := GeometricTransformation[SpikeyT2, TranslationTransform[P]]
MoveSpikeyT2[t_] := Graphics3D[PositionT2[SP2[t]], Axes → True, PlotRange → 9]
```

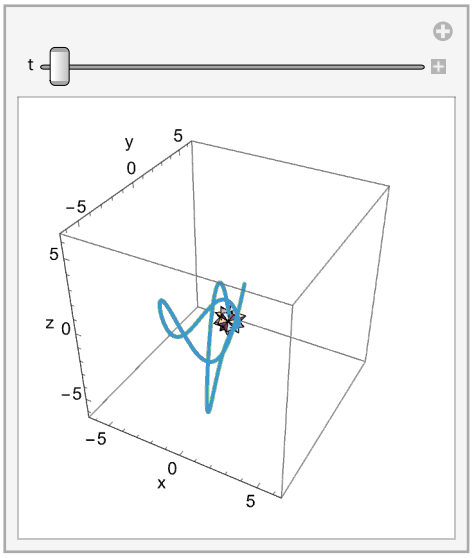
In[13]:=
```
Table[MoveSpikeyT2[t], {t, 0, 3}]
```

Out[13]=



In[14]:=
```
Manipulate[Show[MoveSpikeyT2[t], Traj2, PlotRange → 6.5], {t, -3.5, 3.5}]
```

Out[14]=

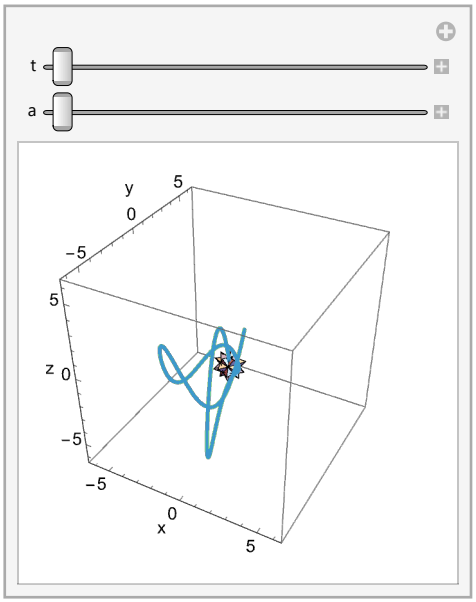


In[15]:=
```
(* Problem 1A *)
```

In[16]:=
```
Position1A[P_, a_] := GeometricTransformation[SpikeyT2,
  Composition[TranslationTransform[P], RotationTransform[a, {0, 0, 1}]]]
MoveSpikey1A[t_, a_] := Graphics3D[Position1A[SP2[t], a], Axes → True, PlotRange → 9]
```

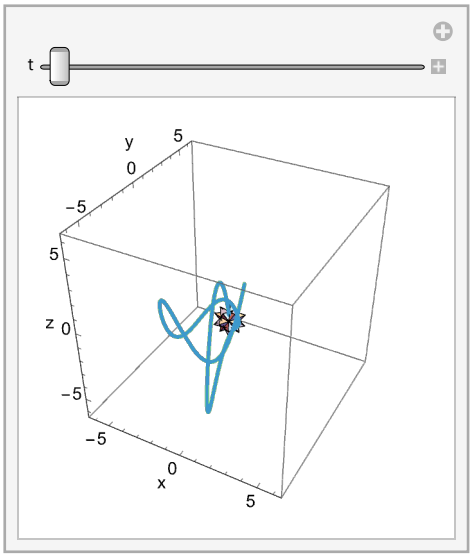
In[18]:=
```
Manipulate[Show[MoveSpikey1A[t, a], Traj2, PlotRange → 6.5], {t, -3.5, 3.5}, {a, 0, 8}]
```

Out[18]=



In[19]:=
```
Manipulate[Show[MoveSpikey1A[t, 5 * t], Traj2, PlotRange → 6.5], {t, -3.5, 3.5}]
```

Out[19]=



In[20]:=
```
SP2[t]
```

Out[20]=

$$\{0.1125\, t\, (-11 + t^2),\ 0.15\, (-12\, t^2 + t^4),\ 0.005\, t\, (-12 + t^2)\, (-9 + t^2)\, (-3 + t^2)\, (-1 + t^2)\}$$

In[21]:=
```
NP = 200
```

Out[21]=

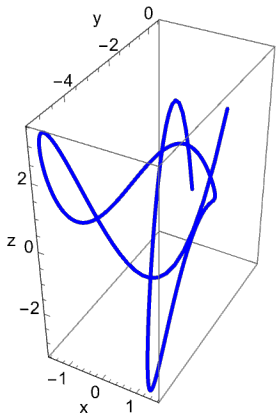
```
200
```

In[22]:=
```
Traj200 = Array[SP2, NP, {-3.5, 3.5}];
```

In[23]:=
```
PTraj200 = Line[Traj200];
```

In[24]:=
```
GTraj200 = Graphics3D[{Blue, Thick, {PTraj200}}, Axes → True]
```

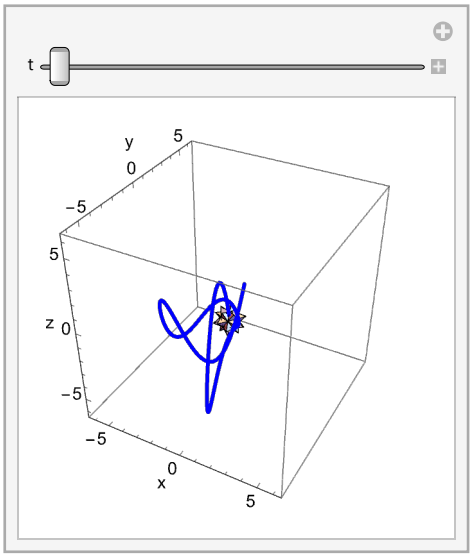
Out[24]=



In[25]:=
```
Position1B[P_, a_] := GeometricTransformation[SpikeyT2,
  Composition[TranslationTransform[P], RotationTransform[a, {0, 0, 1}]]]
MoveSpikey1B[t_, a_] := Graphics3D[Position1B[Traj200[[t]], a], Axes → True, PlotRange → 9]
```

In[27]:=
```
Manipulate[Show[MoveSpikey1B[t, 5 * t], GTraj200, PlotRange → 6.5], {t, 1, 200, 1}]
```

Out[27]=



In[28]:=
```
(* Problem 2 *)
```

In[29]:=
```
t2 = 0; step2 = 0.1;
```

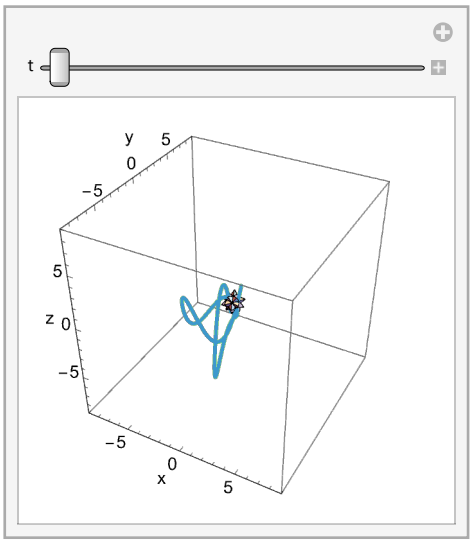
In[30]:=
```
Menu2 = Framed[PopupMenu[Dynamic[menu2], {False → "Stop", True → "Start"}, Background → Red]]
```

Out[30]=



In[31]:=
```
Manip1A = Manipulate[
  If[menu2, step2 = 0.075, step2 = 0];
  If[t2 ≥ 3.5, t2 = -3.5];
  t2 += step2;
  G2 = Show[MoveSpikey1A[t2, t2], Traj2];
  G2, {t, 0, 0}]
```

Out[31]=



In[32]:=
```
(* Problem 3 *)
```

In[33]:=
```
Rect3 = {LightBlue, Rectangle[{0, 0}, {1, 1}]}
```

Out[33]=

{□, Rectangle[{0, 0}, {1, 1}]}

In[34]:=
```
{□, Rectangle[{0, 0}, {1, 1}]}
```

Out[34]=

{□, Rectangle[{0, 0}, {1, 1}]}

In[35]:=
```
R3 = 0.1;
```

In[36]:=
```
D3[x_, y_] := {Blue, Disk[{x, y}, R3]}
```

In[37]:=
```
x3 = 0.5; y3 = 0.5; vx3 = 1; vy3 = Pi / 2; step3 = 0.01;
```

In[38]:=
```
O3[x_, y_] := {Rect3, D3[x, y]}
```

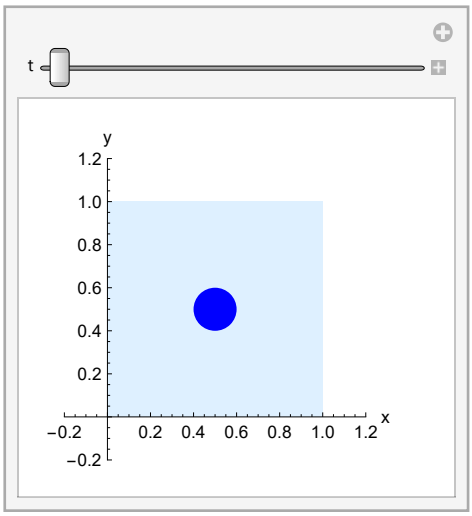
In[39]:=
```
Menu3 = PopupMenu[Dynamic[menu3],
  {False → "Stop", True → "Start"}, MenuStyle → {Orange, "Input"}]
```

Out[39]=

Stop

In[40]:=
```
Manipulate[
 If[menu3, step3 = 0.01, step3 = 0];
 x3 = x3 + vx3 * step3;
 y3 = y3 + vy3 * step3;
 (*If[Abs[x3-1]≤R3||Abs[x3]≤R3,vx3=-vx3];
   If[Abs[y3-1]≤R3||Abs[y3]≤R3,vy3=-vy3];*)
 If[(x3 + R3) ≥ 1 || (x3 - R3) ≤ 0, vx3 = -vx3];
 If[(y3 + R3) ≥ 1 || (y3 - R3) ≤ 0, vy3 = -vy3];
 G3 = Graphics[O3[x3, y3], PlotRange → {{-0.2, 1.2}, {-0.2, 1.2}}, Axes → True];
 G3, {t, 0, 0}]
```

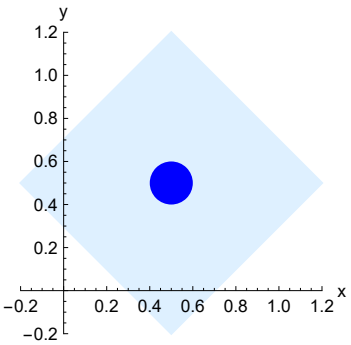
Out[40]=



In[41]:=
```
Rot3B[O_, x_, y_, a_] := GeometricTransformation[O[x, y], RotationTransform[a, {1 / 2, 1 / 2}]]
```

In[42]:=
```
Graphics[Rot3B[O3, 1 / 2, 1 / 2, Pi / 4],
 PlotRange → {{-0.2, 1.2}, {-0.2, 1.2}}, Axes → True, AxesOrigin → {0, 0}]
```

Out[42]=



In[43]:=
```
Rot3B[O_, x_, y_, a_] := GeometricTransformation[O[x, y], RotationTransform[a, {1 / 2, 1 / 2}]]
```

In[44]:=
```
MenuR3B = PopupMenu[Dynamic[menur3],
  {False → "Stop", True → "Start"}, MenuStyle → {Purple, "Input"}]
```

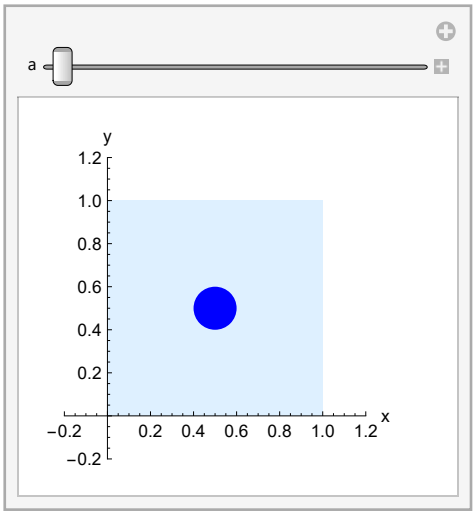
Out[44]=



In[45]:=
```
x3B = 0.5; y3B = 0.5; vx3B = 1; vy3B = Pi / 2; step3B = 0.01; R3 = 0.1; a3B = 0;
```

In[46]:=
```
Manipulate[
 If[menur3, step3B = 0.01, step3B = 0];
 x3B = x3B + vx3B * step3B;
 y3B = y3B + vy3B * step3B;
 If[x3B + R3 ≥ 1 || x3B - R3 ≤ 0, vx3B = -vx3B];
 If[y3B + R3 ≥ 1 || y3B - R3 ≤ 0, vy3B = -vy3B];
 Graphics[Rot3B[O3, x3B, y3B, a],
  PlotRange → {{-0.2, 1.2}, {-0.2, 1.2}}, Axes → True], {a, 0, 1}]
```

Out[46]=



In[47]:=

In[48]:=
```
(* Problem 4 *)
```

In[49]:=
```
Norm[{X, Y}]
```

Out[49]=

$$\sqrt{\text{Abs}[X]^2 + \text{Abs}[Y]^2}$$

In[50]:=
```
Init4[] := Module[{},
  x4 = 0.5;
  y4 = 0.5;
  vx4 = 1;
  vy4 = Pi / 2;
  step4 = 0.1;
  a4 = 0;
  RC4 = 5 / 2;
  RD4 = 0.25;
  ad4 = 0;
  Rad4 = 7 / 2;
  vxold = 0;
  vyold = 0;
  0]
```

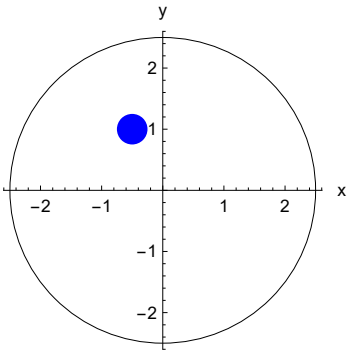
In[51]:=
```
Init4[]
```
Out[51]=
```
0
```

In[52]:=
```
BigCircle4 = Circle[{0, 0}, RC4];
```

In[53]:=
```
D4[x_, y_] := {Blue, Disk[{x, y}, RD4]}
```

In[54]:=
```
O4[x_, y_] := {BigCircle4, D4[x, y]}
```

In[55]:=
```
Graphics[O4[-0.5, 1], Axes → True]
```
Out[55]=



In[56]:=
```
Init4[]
```
Out[56]=
```
0
```

In[57]:=
```
Menu4 = PopupMenu[Dynamic[menu4], {False → "Stop", True → "Start"},
  MenuStyle → {Brown, "Input"}, Background → Yellow]
```
Out[57]=



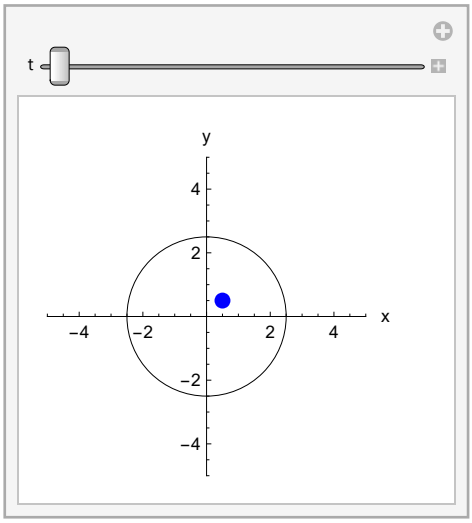
In[58]:=
```
Manip4 = Manipulate[
  If[menu4, step4 = 0.05, step4 = 0];
  vxold = vx4;
  vyold = vy4;

  (* Speed-up if near center *)
  If[(Norm[{x4, y4}])^2 ≤ (RC4 - 2 * RD4)^2 && menu4,
   {vx4 = vx4 * RandomReal[{1, 2}], vy4 = vy4 * RandomReal[{1, 2}]}];

  x4 += vx4 * step4;
  y4 += vy4 * step4;
  vx4 = vxold;
  vy4 = vyold;

  If[(Norm[{x4, y4}])^2 ≥ (RC4 - RD4)^2, {vx4 = -vx4; vy4 = -vy4}];
  GB4 = Graphics[O4[x4, y4], PlotRange → 5, Axes → True];
  GB4, {t, 0, 0}]
```

Out[58]=



In[59]:=
```
Menu5 = PopupMenu[Dynamic[menu5], {False → "Stop", True → "Start"},
  MenuStyle → {Pink, "Input"}, Background → Yellow]
```

Out[59]=



In[60]:=
```
O42[x_, y_, a4_] := GeometricTransformation[O4[x, y],
  Composition[RotationTransform[a4], TranslationTransform[{5 / 2, 0}]]]
```

In[61]:=
```
Manip5 = Manipulate[
    If[menu5, step4 = 0.05, step4 = 0];
    vxold = vx4;
    vyold = vy4;

    (* Speed-up if near center *)
    If[(Norm[{x4, y4}])^2 ≤ (RC4 - 2 * RD4)^2 && menu5,
     {vx4 = vx4 * RandomReal[{1, 2}], vy4 = vy4 * RandomReal[{1, 2}]}];

    x4 += vx4 * step4;
    y4 += vy4 * step4;
    vx4 = vxold;
    vy4 = vyold;

    If[(Norm[{x4, y4}])^2 ≥ (RC4 - RD4)^2, {vx4 = -vx4; vy4 = -vy4}];
    GB4 = Graphics[O42[x4, y4, a4], PlotRange → 5, Axes → True];
    GB4, {a4, 0, 2 Pi}];
```